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On probabilistic prediction and information-theoretic operators

Tracing Shannon's footprints across self-supervised learning

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Abstract

Masked pre-training learns by predicting missing parts of the data, but not all masked prediction tasks are equally informative. We ask whether information-theoretic quantities can identify which observations reduce predictive uncertainty before assimilating them. We introduce predictive information gain (PREDIG), an operator that measures the reduction in uncertainty over target predictions after observing a candidate input. Unlike parameter-space criteria such as BALD, PREDIG evaluates information in the observable predictive space. We analyse PREDIG in Bayesian linear regression and probabilistic principal component analysis (PPCA), moving from a controlled sequential setting to masked image prediction with closed-form Gaussian conditionals. Our results show that PREDIG can support an accept–reject policy, rejecting over 80% of candidate observations in selected settings while preserving predictive performance.

1 Introduction and motivation

Modern artificial intelligence systems are often presented as a story of scale. Larger models, larger datasets, and larger computational budgets have produced systems with remarkable capabilities (Kaplan et al., 2020). Large language models (LLMs) are one of the clearest example: they learn from massive collections of data (Devlin et al., 2019; Brown et al., 2020), and then transfer this predictive structure to a wide range of downstream tasks (Raffel et al., 2023).

At the core of this success lies a way of learning based on a simple idea. A model can learn without external labels by predicting what is missing. In natural language processing (NLP), this may mean predicting hidden tokens from context; in vision, predicting masked patches from the rest visible image. This principle, known as masked pre-training (MPT), has become one of the central mechanisms of significant advances in representation learning (Devlin et al., 2019; He et al., 2021; Moreno-Muñoz et al., 2023).

Yet masking is usually treated in many problems as an heuristic. Parts of the input are *hidden at random*, and the model is trained on the resulting prediction task. This blind, stochastic approach introduces a fundamental inefficiency: the model wastes immense computational budget, time, and hardware resources processing redundant or trivial masks that offer negligible predictive novelty. This works well in practice, but it leaves an important question open: *are all masks equally useful?* Some masked regions may force the model to learn meaningful structure, while others may be almost redundant given what the model already knows. If the goal of learning is to reduce uncertainty, then the choice of what to mask and what to observe should matter. (He et al., 2021)

This question becomes more relevant as models grow in number of parameters. Modern pre-training pipelines process enormous amounts of data, much of which may be repetitive or weakly informative for the current state of the model. The cost is not only computational, but also economic and

environmental (Crawford, 2021). If part of the data is redundant, then the objective should not only be to train on more data, but to quantify which observations are expected to provide new predictive information.

We explore that idea through a probabilistic lens. We study whether predictive uncertainty can be used to decide which data units, patches, or masks are worth processing before the model is updated. The central object is **predictive information gain (PredIG)**: *a measure of how much observing a candidate datapoint, region or mask is expected to reduce uncertainty about a masked target.*

1.1 Background

Across several machine learning paradigms, the central operation is not simply to fit parameters, but to predict unknown quantities from observed ones. This idea appears both in how models are evaluated and in how they are trained. In cross-validation (CV), models are evaluated by leaving a subset of the data out and conditioning on the rest (Bishop, 2006; Fong and Holmes, 2020). In this sense, CV evaluates whether the information learned from the observed data is sufficient to predict unseen observations. In self-supervised learning (SSL) approaches such as MPT (He et al., 2021; Moreno-Muñoz et al., 2023), the learning process involves predicting patches from the context given to the model; and in recent techniques such as joint embedding eredictive rchitecture (JEPA) by Assran et al. (2023), the goal is to learn a target representation from a context representation. Although these methods differ in objective, they share a common structure: learning is about predicting under partial information.

In the supervised setting, this predictive view can be naturally expressed by Bayesian inference. Given a labelled dataset $\mathcal{D} = (x_i, y_i)_{i=1}^{N}$, observations update a posterior distribution over the model parameters, $p(\theta|\mathcal{D})$, but their practical effect is expressed through the posterior predictive distribution (MacKay, 2003a) obtained as:

$$p(y_*|x_*, \mathcal{D}) = \int p(y_*|x_*, \theta)\, p(\theta|\mathcal{D})\, d\theta, \tag{1.1}$$

which integrates parameter uncertainty into predictions about observable quantities. This distribution tell us what the model is able to predict after being conditioned on some data $\mathcal{D}$ and input x_i*.

Information theory provides the quantitative language used in this project to compare uncertainty before and after observing data, which is the reduction in predictive variance. Given a predictive distribution, uncertainty is measured by entropy. The logarithm base sets the information unit, as shown in Table 1. Throughout this work, we use natural logarithms, so information is measured in nats.

Table 1: Units of information measurement (MacKay, 2003a).

Unit	Expression
Bit	$\log_2 p$
Nat	$\log_e p$
Ban	$\log_{10} p$
Deciban	$10 \log_{10} p$

Under this view, information is model-dependent: a data unit is informative when conditioning on it reduces uncertainty about a relevant predictive target. This allows us to assign a ***numerical score*** to a candidate observation interpreted as its expected reduction in predictive uncertainty.

Predictive information as a criterion for redundancy. A central idea behind this work is that not all information contained in data is equally useful for learning. In particular, information that does not improve prediction can be considered redundant for the learning problem. This intuition is closely related to the notion of predictive information introduced by Bialek et al. (2001):

> *"Learning itself can be seen as [...] discover the parameters of an underlying model by trying to compress the information that one sample of N points provides about the next, and in this way we address directly the problem of generalization. [...] The fact that nonpredictive information is useless to the organism suggests that one crucial goal of neural information processing is to separate predictive information from the background."*

This perspective motivates the accept–reject view we would like to adopt. Rather than selecting data because they are merely uncertain or because they are expected to change the parameters of the model, we ask whether an observation reduces uncertainty about relevant missing, future, or target quantities.

The formulation is also consistent with MacKay's view of Bayesian inference (MacKay, 1992a, 2003a). In Bayesian model comparison, the marginal likelihood, or evidence, evaluates how well a model explains the observed data after integrating over parameter uncertainty. This quantity naturally trades off data fit and model complexity, a property often described as Bayesian Occam's razor (MacKay, 2003b). In this project, we use the same predictive spirit at the level of data valuation: if evidence asks which model best explains the observed data, predictive redundancy filtering asks which parts of the data stream are still worth processing.

Guided by these insights, we formulate three core research questions:

RQ-I How can masked pre-training be understood as a predictive protocol built from conditional distributions, and connected to Bayesian notions of marginal likelihood and generalization?

RQ-II Can information-theoretic predictive quantities identify when a candidate observation or batch is informative rather than redundant?

RQ-III To what extent can a predictive accept–reject policy reduce data assimilation while preserving predictive performance?

1.2 Our work and contributions

To address these questions, we implement the predictive information gain criterion as an online accept–reject policy for data assimilation. At each step, the model is represented by its current posterior state over parameters,

$$q_t(\theta) \approx p(\theta \mid \mathcal{D}_t),$$

which induces a posterior predictive distribution over the target quantity T. A candidate batch is assimilated only if its expected reduction of predictive uncertainty, measured in nats, is large enough to justify updating the model.

Pseudo-algorithm 1 Predictive Accept–Reject Criterion

Input: Current dataset $\mathcal{D}_t$, model state $q_t(\theta)$, candidate batch b_t, target quantity T, threshold τ.
Output: Decision $a_t \in \{\text{Accept}, \text{Reject}\}$, updated dataset $\mathcal{D}_{t+1}$, and updated model state $q_{t+1}(\theta)$.
Estimate the current predictive uncertainty about the target T under $q_t(\theta)$.
Estimate how much this uncertainty would be reduced if the candidate batch b_t were assimilated.
Assign a predictive information score s_t to the candidate batch.
if $s_t > \tau$ **then**
    Accept the candidate batch.
    Update the active dataset: $\mathcal{D}_{t+1} = \mathcal{D}_t \cup b_t$.
    Update the model state: $q_{t+1}(\theta) = p(\theta \mid \mathcal{D}_{t+1})$.
else
    Reject the candidate batch as redundant under the current predictive criterion.
    Keep the active dataset unchanged: $\mathcal{D}_{t+1} = \mathcal{D}_t$.
    Keep the model state unchanged: $q_{t+1}(\theta) = q_t(\theta)$.
end if

This sets the technical direction of the present work. The criterion does not value a batch by its intrinsic difficulty alone, nor by the magnitude of a parameter update. Instead, it asks whether conditioning the current probabilistic model on the candidate batch would reduce uncertainty about a relevant target quantity. We first test this idea in tractable linear models, where posterior updates and predictive information gain can be computed analytically. We then move to masked representation learning, where the same predictive principle is used to filter high-dimensional data while preserving predictive performance.

2 Information theory, Bayesian modeling and prediction

Information theory provides a quantitative language for uncertainty (Shannon, 1948). Given a random variable x with probability mass function $p(x)$, the information content associated with observing the outcome x is defined as

$$h(x) = -\log p(x). \tag{2.1}$$

Outcomes that are unlikely under the model carry larger information content, while outcomes that were already expected carry less. Entropy is the average information content under the distribution,

$$H(x) = \mathbb{E}_{p(x)}[-\log p(x)]. \tag{2.2}$$

This notion is useful for learning because probabilistic models do not only produce point estimates: they define distributions over possible outcomes. A broad learning principle can then be stated as follows: observations are useful when they change what the model is able to predict, and redundant when they leave the relevant predictive uncertainty essentially unchanged.

In probabilistic machine learning, the Bayesian framework treats learning as the update of a probability distribution over unknown quantities. Under this paradigm, prediction is not made from a single fitted parameter value, but from the posterior distribution induced by the observed data (MacKay, 1992a; Bishop, 2006; Murphy, 2012). Given a dataset $\mathcal{D}_N = \{(x_i, y_i)\}_{i=1}^N$, the central predictive objective is the posterior predictive distribution, see equation 1.1 where $p(y_* \mid x_*, \theta)$ is the prediction made by the model for a fixed parameter value, and $p(\theta \mid \mathcal{D})$, the posterior weight assigned to that parameter value after observing the data. Thus, the posterior predictive distribution averages predictions over all plausible parameter values instead of relying on a single estimate of θ.

This predictive view also gives a natural interpretation of the marginal likelihood, or *evidence*. For a supervised model, the evidence can be written as:

$$p(\mathcal{D}_N) = \int p(\mathcal{D}_N|\theta)p(\theta)\, d\theta. \tag{2.3}$$

Here, $p(\mathcal{D}_N|\theta)$ is the likelihood of the observed outputs under a fixed parameter value, while $p(\theta)$ is the prior distribution over parameters. The marginal likelihood therefore evaluates how well the model explains the data after averaging over the parameter values allowed by the prior.

Using the chain rule of probability, the same quantity can be written as a sum of conditional log-probabilities:

$$\log p(\mathcal{D}_N) = \sum_{i=1}^{N} \log p(y_i|x_i, \mathcal{D}_{1:i-1}), \tag{2.4}$$

where $\mathcal{D}_{1:i-1} = \{(x_j, y_j)\}_{j=1}^{i-1}$ denotes the history of observations available prior to step i.

Predictive sequential view Equation 2.4 is the predictive-sequential view of marginal likelihood (Dawid, 1984). Rather than evaluating a model only after it has been fitted to the full dataset, each observation is scored before being incorporated into the training set. In this view, the marginal likelihood acts as a cumulative predictive score: a model is rewarded when it assigns high probability to observations before seeing them, while unnecessary flexibility is naturally penalized.

Building upon this predictive-sequential view, selecting data using information-theoretic objectives has a rich history spanning from classical foundations (MacKay, 1992b) to deep Bayesian active learning frameworks (Gal et al., 2017; Kirsch et al., 2023). Recent advancements further emphasize that not all observations are equally valuable for improving future predictions. As argued in prediction-oriented Bayesian active learning (Smith et al., 2023a), the primary goal should be selecting data that explicitly improves performance on a target distribution, rather than merely reducing parameter uncertainty.

A closely related perspective is explored by Kirsch (2023), who notes that examples should not be valued solely by their current loss or gradient norm. While high-loss points might appear informative, they can often be inherently noisy, ambiguous, or simply irrelevant to future generalization. Instead, the relevant question is strictly predictive: *how much would training on a candidate point reduce the error on unseen data?* This leads to the reducible holdout loss (RHO-Loss), an acquisition function

that prioritizes examples whose loss is genuinely reducible, rather than examples that are merely hard to predict:

$$\mathrm{RHO}(x, y) = H[y|x, \mathcal{D}_{\mathrm{train}}] - H[y|x, \mathcal{D}_{\mathrm{eval}}]. \tag{2.5}$$

Here, $H[\cdot]$ denotes the predictive entropy, which in standard classification tasks would be equivalent to the cross-entropy loss. The RHO-Loss is also computed as the difference between two terms. The first term, $H[y|x, \mathcal{D}^{\mathrm{train}}]$, represents the uncertainty (or loss) of the current model trained on the available dataset. The second term, $H[y|x, \mathcal{D}^{\mathrm{eval}}]$, represents the inherent or irreducible uncertainty of that point, estimated using an oracle or *reference model* trained on a clean, held-out evaluation set. By subtracting the latter from the former, RHO-Loss isolates the portion of the error that the current model can actually learn to overcome. In this sense, useful data are those that are not yet learned, mathematically learnable, and relevant for generalization.

This perspective shifts the Bayesian predictive view from model selection to data selection. The question is not only which model best explains the observed data, but also which observations are still worth processing. We study data usefulness through its effect on future predictive uncertainty, rather than through training error alone. This connects Bayesian modeling with information theory: if entropy measures predictive uncertainty, then learning can be studied through how observations change the uncertainty of future, missing, or target quantities.

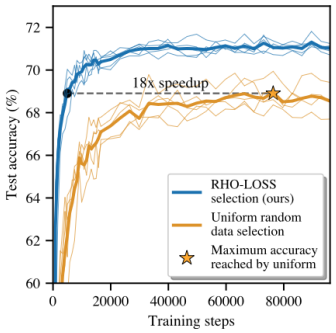


Figure 1: Example of RHO-LOSS Speedup on large-scale classification of web-scraped data. Kirsch (2023)

3 Self-supervised learning and SOTA

In Bayesian modeling, the marginal likelihood is commonly used as a measure of generalization ability, since it evaluates how well a model explains the data after integrating over its latent hypotheses (Tenebaum and Griffiths, 2001; MacKay, 2003a). This makes a question emerge: *if masked pretraining generalizes well in practice, can its success be interpreted as evidence maximization?*

As a reference, masked pre-training (MPT) is one of the most early forms of self-supervised learning used in BERT by Devlin et al. (2019). Its basic mechanism is simple: given an observation $x = (x_1, x_2, \ldots, x_D)^T$, a subset of tokens is masked, and the model aims to predict the missing components from the remaining context. If M denotes the set of masked indices and $R = \{1, 2, \ldots, D\} \setminus M$ the observed context, the standard MPT objective can be written as

$$\log p_\theta(x_{\mathcal{M}} \mid x_{\mathcal{R}}) \approx \sum_{j \in \mathcal{M}} \log p_\theta(x_j \mid x_{\mathcal{R}}). \tag{3.1}$$

or, in practice, as a sum of predictive log-probabilities.

This objective fits within the predictive view introduced earlier: learning is organized around predicting unobserved quantities from observed ones. The connection is made precise by (Moreno-Muñoz et al., 2023), who show that *masked pretraining with an appropriate cumulative scoring function corresponds to maximizing the model's marginal likelihood*. The key idea is that the log-probability of an observation can be decomposed with chain rule into a sequence of conditional predictive terms. Thus, MPT samples random masks and learns from many conditional views of the same observation.

In the predictive sequential formulation, the marginal likelihood scores how well a model predicts each new observation given previous observations (Dawid, 1984; MacKay, 2003a). In masked pre-training, the model is instead scored by how well it predicts missing components given observed ones within the same sample (Devlin et al., 2019; He et al., 2021). Thus, both perspectives evaluate learning through conditional prediction under partial information.

This connection is central to our work. Masked pre-training already defines a conditional prediction problem: visible tokens act as context, and masked tokens act as targets. PredIG uses this structure to measure how much a candidate context reduces predictive uncertainty about the missing components. Therefore, the MPT formulation provides the bridge between Bayesian prediction, self-supervised learning, and the information-theoretic data-valuation criterion studied.

3.1 State of the art and the predictive view

The intersection of probabilistic machine learning and information theory provides a principled framework for active learning: while the first treats learning as the sequential update of probability distributions over unknown quantities, the latter provides the mathematical currency as entropy or mutual information to quantify how much a new observation will alter those distributions.

Several authors have studied information-theoretic formulations of active learning. BALD introduced Bayesian Active Learning by Disagreement, where candidate observations are selected according to their mutual information with model parameters (Houlsby et al., 2011). To extend this to co-dependent acquisitions, Kirsch et al. (2019) developed BATCHBALD, which scores candidate groups jointly to prevent the selection of mutually redundant data. Later works such as EPIG shifted these ideas toward predictive objectives that explicitly account for test distributions and predictive uncertainty rather than parameter space (Smith et al., 2023b). While these approaches successfully quantify the informational value of observations, they are primarily formulated within active acquisition frameworks rather than a unified, continuous observable space.

In parallel, modern compute-efficient learning methods have explored selective computation through dataset pruning, routing, curriculum learning, and prioritized training (Mindermann et al., 2022). Notably, Paul et al. (2023) demonstrated that influential examples can be identified early in training via empirical metrics like the error L2 norm (EL2N), allowing significant data fractions to be safely discarded. These co-related approaches acknowledge that not all observations contribute equally to learning; yet, most existing methods rely on heuristic criteria, specific architectural mechanisms, or empirical loss-based scores rather than an explicit, principled measure of predictive information gain.

The predictive view principles have also become central in self-supervised learning. Methods based on masked prediction of masked tokens, representation learning, and joint-embedding predictive architectures implicitly optimize predictive objectives by reconstructing or anticipating missing information from context (He et al., 2021; Assran et al., 2023). Moreno-Muñoz et al. (2023) have also explored connections between masked pre-training and marginal likelihood from Bayesian perspective. However, these methods rarely admit an explicit probabilistic interpretation in terms of predictive uncertainty reduction and information gain in observable space.

4 Linear-in-the-parameters models and information-theoretic lens

To study predictive information dynamics in a controlled setting, we first consider a linear-in-the-parameters models to preserve the analytic tractability. Bayesian linear regression (BLR) is a useful scenario because objects like posterior predictive distribution, marginal likelihood and conditional entropies quantities are available in closed form. This automatically removes optimization constraints such as stochastic gradient noise to set a local minima solution, and allows us to isolate whether predictive criteria can distinguish informative observations from redundant ones.

Bayesian linear regression. We consider a dataset $\mathcal{D}_N = \{(x_i, y_i)\}_{i=1}^N$ with inputs $x_i \in \mathbb{R}$ and targets $y_i \in \mathbb{R}$, generated from a polynomial basis model with additive Gaussian noise $y_i = \phi(x_i)^\top w + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$ and $\phi(x) \in \mathbb{R}^{p+1}$ denotes a polynomial feature map. Placing an isotropic Gaussian prior over the weights, $p(\theta) = \mathcal{N}(\theta \mid 0, \alpha^{-1} I)$, ensures that the parameter posterior remains analytically tractable. Specifically, given the current dataset, the posterior distribution is Gaussian, $p(\theta \mid \mathcal{D}_N) = \mathcal{N}(\theta \mid m_N, S_N)$, with its parameters defined as:

$$S_N^{-1} = \alpha I + \sigma_n^{-2} \Phi^\top \Phi, \qquad m_N = \sigma_n^{-2} S_N \Phi^\top y, \tag{4.1}$$

where $\Phi \in \mathbb{R}^{N \times (p+1)}$ is the design matrix and $y \in \mathbb{R}^N$ is the vector of targets. This closed-form tractability allows us to compute exact posterior updates without relying on stochastic optimization or variational approximations.

Posterior predictive distribution. The predictive distribution is obtained by integrating out the posterior uncertainty in the weights:

$$\begin{aligned} p(y_* \mid x_*, \mathcal{D}_N) &= \int p(y_* \mid x_*, \theta) p(\theta \mid \mathcal{D}_N)\, d\theta = \\ &= \mathcal{N}\left(\phi(x_*)^\top m_N,\, \sigma_n^2 + \phi(x_*)^\top S_N \phi(x_*)\right). \end{aligned} \tag{4.2}$$

Here, $\phi(x_*)$ is the feature representation of the test input, while m_N and S_N are the posterior mean and covariance of the weights after observing $\mathcal{D}_N$. The predictive mean $\phi(x_*)^\top m_N$ gives the expected target value, whereas the predictive variance quantifies the remaining uncertainty about y_*.

The variance separates two sources of uncertainty: the observation noise σ_n^2, which is irreducible, and the projected posterior covariance $\phi(x_*)^\top S_N \phi(x_*)$, which decreases as informative data are observed (MacKay, 1992a; Bishop, 2006; Murphy, 2012). This posterior predictive distribution is the object whose uncertainty we will later measure with entropy when defining predictive information gain.

4.1 Information-theoretic operators

To view Bayesian learning through an information-theoretic lens is to treat knowledge not merely as an abstract state of belief, but as a quantifiable currency. Information theory provides the rigorous mathematical tool needed to measure the exact volume of uncertainty that is dissolved when a model encounters new data. In this continuous probabilistic learning landscape, learning can be seen as a measurable flow of information, naturally weighed in nats, where the natural logarithm scales the physical degree of "surprise" or uncertainty reduction.

Classical approaches of the information-theoretic criteria in Bayesian learning usually operate in parameter space of θ: a point is informative if it reduces uncertainty about the unknown parameters. For instance, this is the view underlying Bayesian experimental design (BED) from Rainforth et al. (2023) and BALD-style active learning (Houlsby et al., 2011). Our objective is thus different: we aim to be able to detect which observations still improve predictive knowledge and which ones are redundant. This directly connects to **RQ-II**, which asks whether information-theoretic predictive quantities can identify when a candidate observation or batch is informative rather than redundant.

Information gain in parameter space A standard approach based on the parameter-space criterion is bayesian active learning by disagreement (Houlsby et al., 2011) or BALD, which measures the expected information gain (EIG) about the model parameters. For a candidate input x, BALD is

$$I(\theta; y \mid x, \mathcal{D}_N) = H[y \mid x, \mathcal{D}_N] - \mathbb{E}_{p(\theta \mid \mathcal{D}_N)} \left[H[y \mid x, \theta] \right]. \tag{4.3}$$

For the BLR model, this reduces to the tractable expression

$$I(\theta; y \mid x, \mathcal{D}_N) = \frac{1}{2} \log \left(1 + \sigma_n^{-2} \phi(x)^\top S_N \phi(x) \right). \tag{4.4}$$

For a candidate batch B with design matrix Φ_B, the corresponding batch score is

$$I(\theta; y_B \mid X_B, \mathcal{D}_N) = \frac{1}{2} \log \left| I_B + \sigma_n^{-2} \Phi_B S_N \Phi_B^\top \right|. \tag{4.5}$$

This criterion exclusively selects observations that are expected to reduce uncertainty about the underlying model parameters θ.

Figure 2 shows this mechanism qualitatively for Bayesian linear regression with a fourth-degree polynomial basis. Each row corresponds to a different posterior state. The left panel shows the marginal posterior over the first two weights, the middle panel shows posterior predictive samples together with the predictive mean, and the right panel shows the BALD score over candidate inputs.

With only one observation, the posterior over parameters is still broad and many polynomial functions remain plausible. As a consequence, the predictive distribution spreads widely away from the observed point, and the BALD score is high in regions where a new observation would strongly constrain the parameter posterior. After four observations, the predictive samples become more structured, and the BALD values decrease. After six observations, most of the uncertainty in the observed region has been removed: the posterior is much more concentrated, the predictive functions agree more closely, and the BALD curve is lower across the input domain.

This monotonic decrease is not only visual. In BLR, assimilating new observations reduces the posterior covariance in the positive semi-definite sense. Hence, for any fixed candidate input x, the projected uncertainty $\phi(x)^\top S_N \phi(x)$ cannot increase as more data are observed, and therefore the BALD score also decreases. The remaining peaks appear mainly in extrapolation regions, where the polynomial features still project onto uncertain directions of the parameter posterior.

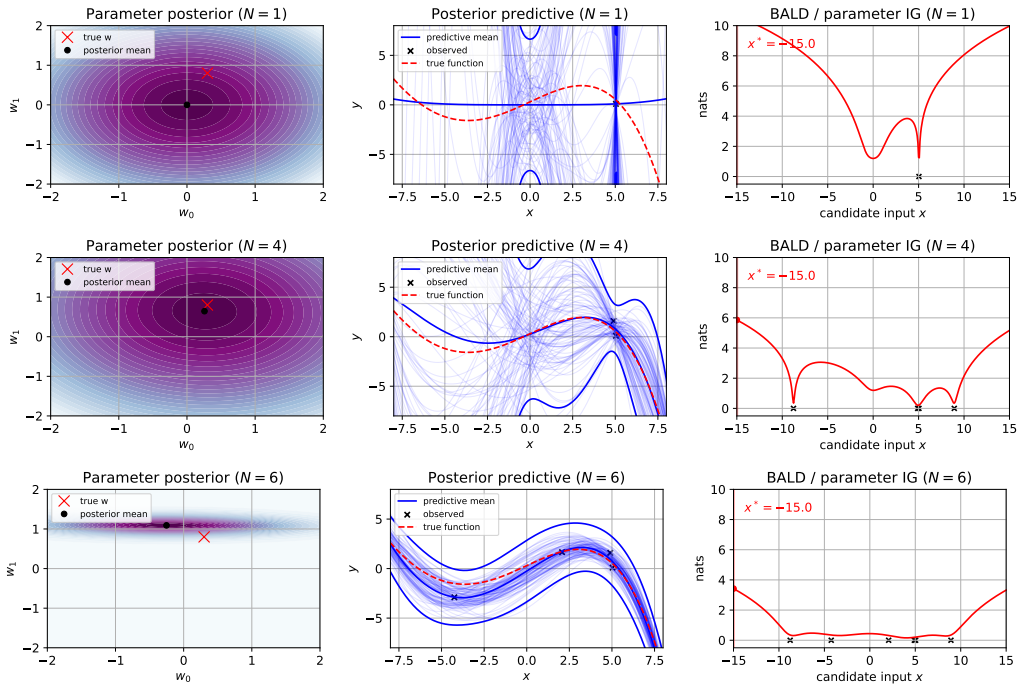


Figure 2: **Parameter-space information gain in Bayesian linear regression.** BALD scores candidate inputs by their expected reduction of posterior uncertainty about the model parameters. Here the model uses a fourth-degree polynomial basis. As observations are assimilated, the posterior covariance contracts, the posterior predictive distribution becomes more concentrated, and the BALD score decreases across the input domain.

This parameter-space view is useful because it makes information gain analytically transparent. However, it also reveals the limitation of the criterion for our objective. BALD values an observation because it contracts uncertainty about θ, not because it is necessarily the most useful observation for a particular predictive target. In simple BLR these two notions are often related, but they are conceptually different. This distinction becomes especially important in overparameterized models, where many different parameter configurations may induce similar predictive behaviour. In such settings, large changes in parameter space do not necessarily correspond to meaningful changes in the predictions (Mindermann et al., 2022).

This motivates a predictive alternative: instead of asking how much a candidate observation tells us about the parameters, we ask how much it reduces uncertainty over the quantities we actually want to predict. The predictive route therefore decouples data valuation from the internal geometry of the parameter space. Its main requirement is the definition of a target x_* over which predictive uncertainty is measured. Once this target is specified, information gain can be aligned directly with predictive utility rather than parameter identification.

Predictive information gain. We are now interested in uncertainty over the masked quantities, not parameter uncertainty itself. Let x_* be a fixed collection of target inputs and Y_* the corresponding predictive variables. Since x_* remains fixed, we omit it from the notation below.

For a single candidate input x, PredIG is the reduction in predictive uncertainty induced by x:

$$\mathrm{PredIG}(x) = \underbrace{H_{\mathrm{before}}[y_*]}_{\text{before } x} - \underbrace{H_{\mathrm{after}}[y_*|x]}_{\text{after } x}, \tag{4.6}$$

where $H[\cdot]$ denotes the differential entropy of the predictive distribution Cover and Thomas (2006); a short review of conditional entropy and its relation to information gain is provided in Appendix A. Specifically, the first term $H_{\mathrm{before}}[y_*]$ represents the current predictive uncertainty over the fixed target set given the existing dataset $\mathcal{D}_N$. The second term $H_{\mathrm{after}}[y_*|x]$ denotes the expected remaining uncertainty over the targets after updating the model with the new candidate input x and

its corresponding latent or observed label. In BLR, this reduction is determined directly by the transformation of the predictive covariance:

$$\mathrm{PredIG}(x) = \frac{1}{2} \log \left(\frac{|\Sigma_{*,N}|}{|\Sigma_{*,N+1}(x)|} \right), \tag{4.7}$$

where

$$\Sigma_{*,N} = \sigma_n^2 I + \Phi_* S_N \Phi_*^\top \tag{4.8}$$

is the predictive covariance before the update, while

$$\Sigma_{*,N+1}(x) = \sigma_n^2 I + \Phi_* S_{N+1}(x) \Phi_*^\top \tag{4.9}$$

is the predictive covariance after the update induced by x, with

$$S_{N+1}(x) = \left(S_N^{-1} + \sigma_n^{-2} \phi(x) \phi(x)^\top \right)^{-1}. \tag{4.10}$$

The same criterion extends directly to a candidate batch B containing M inputs. Its predictive information gain is

$$\mathrm{PredIG}(B) = \underbrace{H_{\mathrm{before}}[Y_*]}_{\text{before } B} - \underbrace{H_{\mathrm{after}}[Y_*; B]}_{\text{after } B}. \tag{4.11}$$

For BLR, the corresponding batch score is

$$\mathrm{PredIG}(B) = \frac{1}{2} \log \left(\frac{|\Sigma_{*,N}|}{|\Sigma_{*,N+M}(B)|} \right). \tag{4.12}$$

Unlike BALD, which measures information about θ, PredIG measures the reduction in uncertainty over the fixed target predictions Y_*. In BLR, these covariance updates depend only on the candidate inputs, allowing both scores to be computed before observing their target values.

4.2 Putting things together: empirical results

A sequential view of predictive information gain. Before exploring use of the PredIG operator as an accept–reject policy, it is useful to inspect how predictive information behaves in a simple sequential Bayesian update. We consider Bayesian linear regression and visualize each update from three complementary perspectives: the posterior distribution over the first two parameters, the posterior predictive distribution in output space, and the local reduction in predictive uncertainty induced by the newly observed point.

Figure 3 shows the initial stage of learning. Before observing any data, the posterior over parameters is broad and the posterior predictive distribution contains many plausible functions. Since the model has not yet constrained the relation between input and output, predictive uncertainty is high over large parts of the domain. After the first observation, uncertainty is reduced around the observed region, and the PredIG curve becomes structured: candidate inputs now receive different values depending on how much they are expected to reduce uncertainty over the fixed target set.

The next stage shows that predictive informativeness is not a static property of an input location. Figure 4 compares the model after five and six observations. After five observations, several regions close to already observed points have low PredIG, because the model can already predict them with relatively small uncertainty. However, after the sixth observation, the posterior predictive state changes: the model becomes more confident in some regions, but the remaining uncertainty is redistributed across the input domain. As a result, a region that previously had low predictive value can become informative again if the new posterior reveals that observing there would reduce uncertainty over the target set.

This behaviour is important for our interpretation of the operator. PredIG is recomputed after each Bayesian update and depends on the current predictive distribution. In a polynomial model, observations in one part of the domain can affect uncertainty elsewhere because the basis functions couple distant input regions. Therefore, the score can decrease in already explained regions while increasing in regions that become relevant under the updated predictive state.

Finally, once many observations have been assimilated, the posterior predictive distribution becomes stable in the data-supported region. Figure 5 shows the model after 199 observations. The posterior

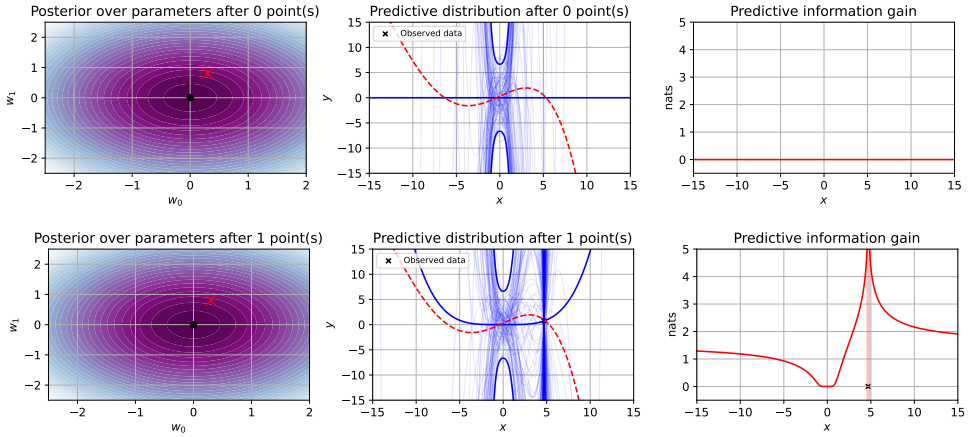


Figure 3: Early evolution of predictive information in Bayesian linear regression. Each row corresponds to one posterior state. **Left**: posterior distribution over the first two parameters, with the red cross denoting the true parameter values and the black point denoting the posterior mean. **Middle**: posterior predictive distribution in output space, where blue curves are posterior samples, the solid blue line indicates the predictive mean, and black crosses are observed data. **Right**: predictive information gain in nats over candidate inputs. Initially, most of the domain is uncertain. After the first observation, PredIG becomes concentrated in regions where an additional observation would most reduce predictive uncertainty over the target set.

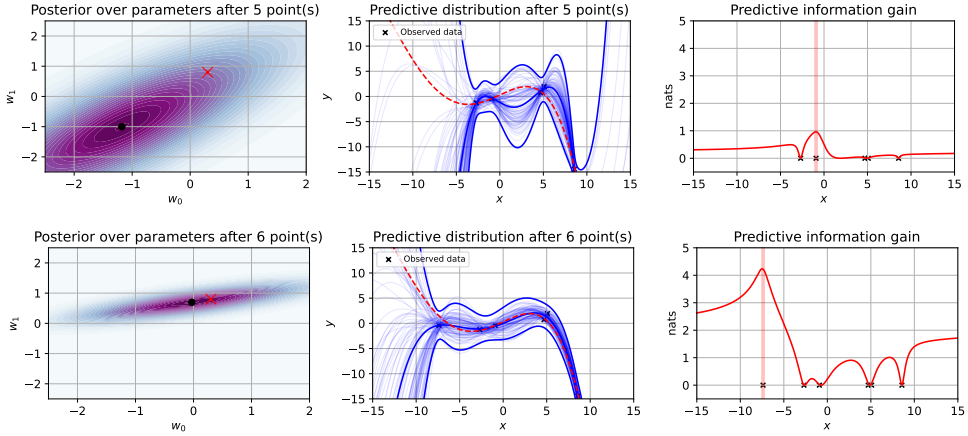


Figure 4: Behaviour of predictive information. After five observations, PredIG is low near regions that are already well constrained by the current posterior predictive distribution. After the sixth observation, the predictive state changes and the information landscape is reshaped. Consequently, regions that previously had low score may become informative again if observing them is now expected to reduce uncertainty over the target predictions. This illustrates that PredIG is a dynamic, posterior-dependent quantity rather than a fixed score attached to each input location.

over parameters is highly concentrated, and the predictive samples agree closely across the interval where data have been observed. In this region, PredIG is almost zero: additional observations no longer reduce predictive uncertainty in a meaningful way, and are therefore redundant from the perspective of the target predictions.

The remaining non-zero PredIG is concentrated near extrapolation regions, where the model has little or no data support. This late-stage behaviour is precisely what makes the operator useful for our objective. As the model becomes confident in a region, repeated observations from that region naturally lose predictive value. At the same time, PredIG can still identify regions where uncertainty remains and where new observations are expected to improve the model's predictive state.

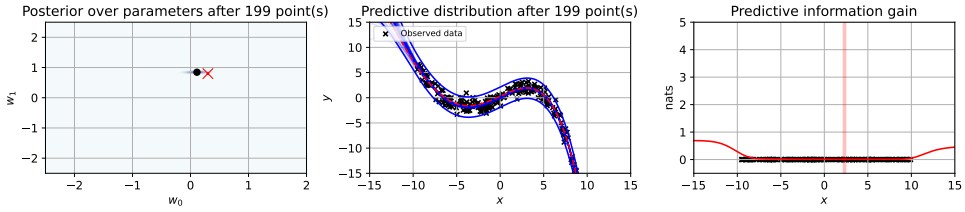


Figure 5: Predictive information after 199 observations. The posterior over parameters is now highly concentrated and the predictive distribution has stabilized in the data-supported region. PredIG is close to zero where observations are dense, indicating that additional data there are mostly redundant. Non-zero information gain remains mainly near extrapolation regions, where predictive uncertainty has not been fully removed.

Evolution of the PredIG operator. Furthermore, we also inspect whether it behaves as expected. The score should be high when a candidate batch can reduce predictive uncertainty over the target grid X_*, and should decrease as the posterior predictive distribution becomes more certain.

Figure 6 shows this behaviour for an ordered stream of batches in the one-dimensional BLR setting. Each column corresponds to a different posterior state after assimilating 0, 20, and 110 batches. The top row shows the posterior distribution over parameters, the middle row shows the posterior predictive distribution in input space, and the bottom row shows the PredIG score assigned to candidate batch locations. Early in the stream, uncertainty is high and PredIG assigns large values across broad regions. As observations are assimilated sequentially, predictive uncertainty contracts first in the observed region, and PredIG concentrates on candidate locations that can still reduce uncertainty over the target grid. Once the stream has covered the relevant input domain, the PredIG landscape collapses toward zero, indicating that new candidate batches are increasingly redundant with respect to the current predictive state.

Effect of batch arrival order. Figure 7 repeats the same diagnostic with unordered batch arrivals. The columns now correspond to 0, 1, and 110 assimilated batches, highlighting the effect of broad early coverage. After a single unordered batch, observations already span multiple regions of the input domain, so predictive uncertainty contracts more globally than in the ordered stream. Consequently, the PredIG landscape flattens much faster. This shows that PredIG is not determined only by the number of observed batches, but also by their coverage relative to the target grid.

Together, these diagnostics suggest that PredIG captures both redundancy and coverage in predictive space. A batch is informative when it is expected to reduce uncertainty about future observable quantities Y_*, not simply when it changes the parameter posterior.

PredIG-based redundancy criterion. We next use the same PredIG operator as an online accept–reject rule for incoming batches. The policy, denoted LIGHT-REJECT, computes the expected predictive information score before assimilating a batch and updates the BLR posterior only when this score exceeds a fixed threshold $\tau = 0.1$. Figure 8 shows a representative run, while Table 2 reports the multi-run statistics over 30 independent streams. The filtered policy rejects most of the stream, using only about 17% of the batches, but remains close to FULL assimilation in RMSE, NLL, and cumulative predictive entropy reduction. The small NLL degradation indicates that assimilating all data still gives slightly better calibrated predictive distributions, but the effect is minor compared with the reduction in posterior updates. This supports the interpretation of PredIG as a redundancy policy: after the posterior has captured the predictive structure of the target distribution, many later observations are redundant for prediction. This supports the interpretation of PredIG as a redundancy policy: after the posterior has captured the predictive structure of the target distribution, many later observations are redundant for prediction. **RQ-III.** These results provide empirical evidence that a predictive accept–reject policy can reduce the amount of assimilated data while largely preserving predictive performance.

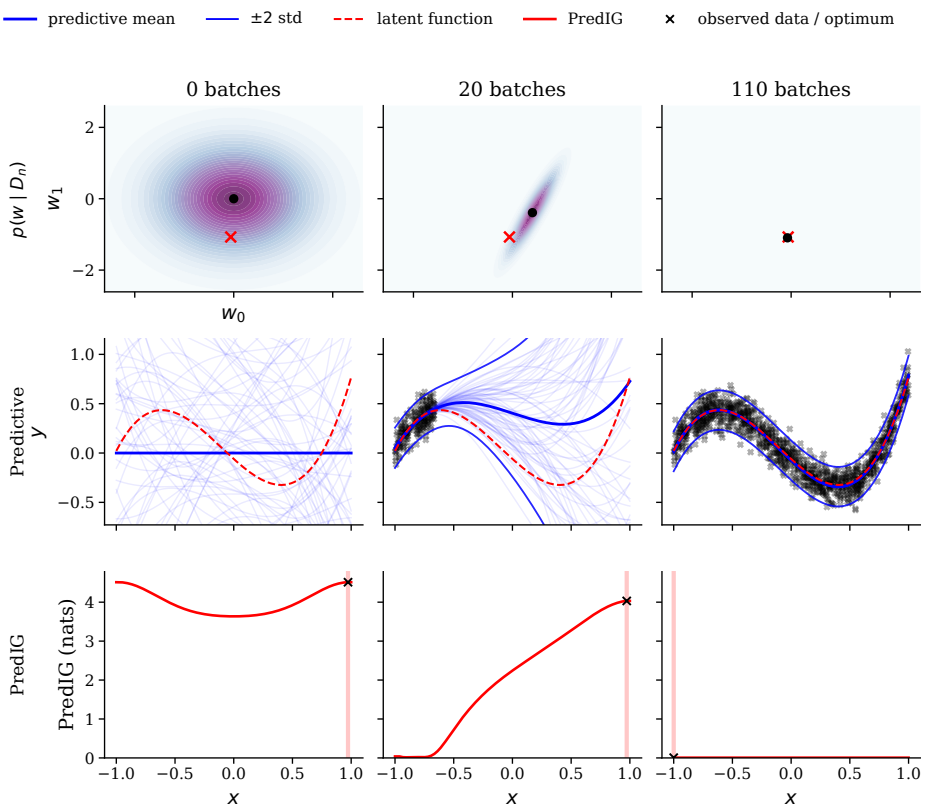


Figure 6: **Evolution of parameter uncertainty, predictive uncertainty, and PredIG under an ordered batch stream.** Each column shows the model state after assimilating 0, 20, and 110 batches. **Top row:** posterior distribution over the BLR parameters, with the red marker indicating the latent parameter configuration and the black marker indicating the current posterior optimum. **Middle row:** posterior predictive distribution, including posterior function samples, predictive mean, ± 2 standard deviation band, latent function, and observed data. **Bottom row:** PredIG landscape over candidate batch locations. The selected candidate is marked at the maximum of the score. As data are assimilated, posterior and predictive uncertainty contract, and the PredIG score decays toward zero, indicating that new candidate batches become increasingly redundant.

5 Masked pre-training via representation learning

Moving beyond scalar regression, we now address high-dimensional structured data. In this regime, representation learning aims to uncover an unobserved, lower-dimensional latent space that explains the underlying generative process of the observations. A standard approach to achieve this without external labels is through self-supervised searning (SSL), specifically using a technique known as masked pre-training (MPT).

Masked pre-training as predictive learning. Let $x \in \mathbb{R}^D$ denote an observation. A mask $\mathcal{M} \subset \{1, \dots, D\}$ selects the hidden coordinates $x_{\mathcal{M}}$, while the complement

$$\mathcal{R} = \{1, \dots, D\} \setminus \mathcal{M}$$

defines the observed context $x_{\mathcal{R}}$. Masked pre-training learns from the conditional prediction problem of reconstructing $x_{\mathcal{M}}$ from $x_{\mathcal{R}}$. A masked prediction task is scored by

$$\log p_\theta(x_{\mathcal{M}} \mid x_{\mathcal{R}}) \approx \sum_{j \in \mathcal{M}} \log p_\theta(x_j \mid x_{\mathcal{R}}). \tag{5.1}$$

The approximation corresponds to scoring the masked coordinates through marginal conditional predictions. This is the predictive structure behind masked language modeling and masked image modeling: the model learns representations by exploiting statistical dependencies between observed and hidden dimensions (Devlin et al., 2019; He et al., 2021).

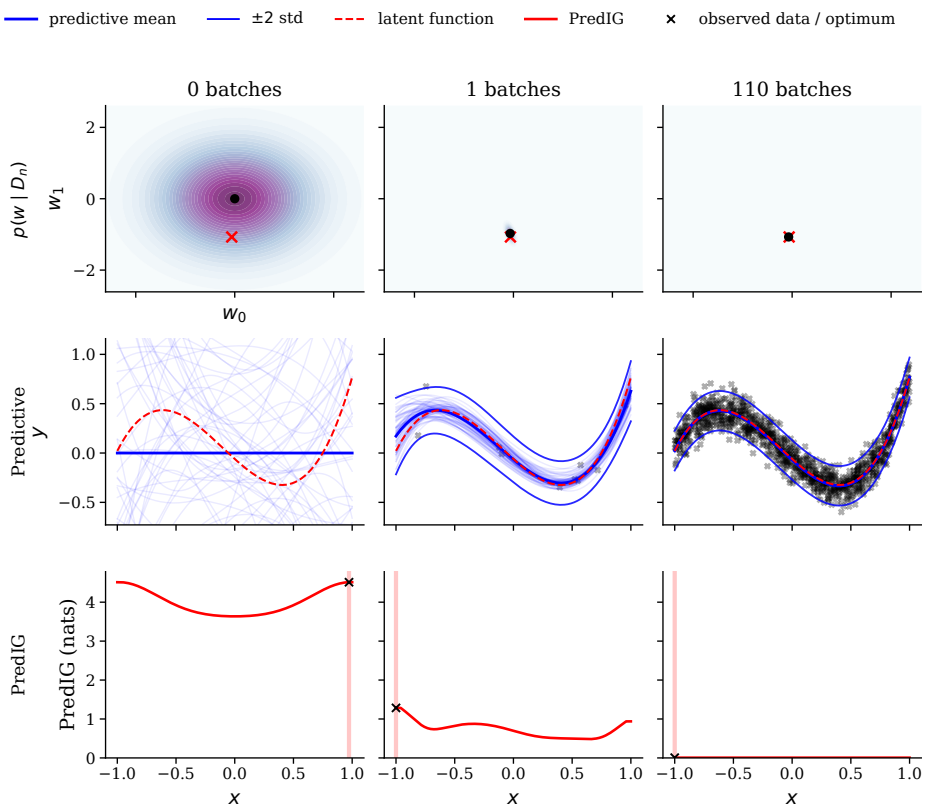


Figure 7: **Evolution of PredIG under unordered batch arrivals.** Broader early coverage reduces predictive uncertainty more globally, causing the PredIG landscape to decay faster than in the ordered stream. Scores are reported in nats.

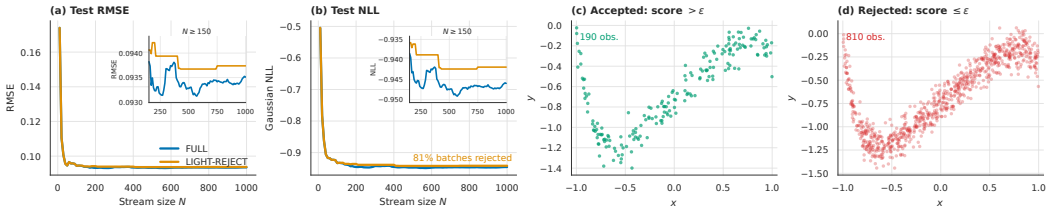


Figure 8: **PredIG-based filtering preserves predictive accuracy while rejecting redundant observations.** We compare FULL assimilation against LIGHT-REJECT in BLR with polynomial degree $p_{\text{active}} = 7$ and threshold $\epsilon = 0.1$. **(a)** Test RMSE for a representative run, with an inset zooming into the behaviour after the initial adaptation phase. **(b)** Gaussian test NLL for the same run, again with a post-transient zoom. **(c)** Observations accepted by the PredIG gate. **(d)** Observations rejected as predictively redundant.

In standard MPT, masks are usually sampled at random. This provides broad coverage over many prediction tasks, but it does not distinguish between masks that are informative and masks that are redundant under the current model. Since the number of possible masks grows combinatorially with the input dimension, the masking distribution determines which conditional prediction tasks the model actually sees during training.

Moreno-Muñoz et al. (2023) showed that, under cumulative scoring over masks, masked pre-training is connected to the model marginal likelihood. This gives MPT a Bayesian interpretation: learning from partial observations can be understood as evaluating how well a probabilistic model explains the full observation through its conditional distributions. We use this perspective to ask a selection

Table 2: **Multi-run summary of PredIG-based redundancy filtering.** Results are mean ± standard deviation over 30 independent streams with $N = 1000$, batch size 10, true polynomial degree $p = 7$, manager degree $p_{\text{active}} = 7$, and threshold $\epsilon = 0.1$.

Metric	FULL	LIGHT-REJECT	Change / fraction
Predictive performance			
Test RMSE ↓	0.1018 ± 0.0039	0.1033 ± 0.0041	$+0.0015 \pm 0.0016$
Test Gaussian NLL ↓	-0.865 ± 0.040	-0.851 ± 0.041	$+0.014 \pm 0.016$
Predictive entropy reduction ↑	2.562 ± 0.000	2.547 ± 0.001	-0.015 ± 0.001
Filtering behaviour			
Accepted batches	100	16.9 ± 1.3	$16.9 \pm 1.3\%$
Rejected batches	0	83.1 ± 1.3	$83.1 \pm 1.3\%$
Accepted observations	1000	169.3 ± 12.8	$16.9 \pm 1.3\%$
Rejected observations	0	830.7 ± 12.8	$83.1 \pm 1.3\%$

question: *before processing a candidate patch or mask, can we estimate how much it is expected to reduce predictive uncertainty?*

Probabilistic PCA for masked prediction. While modern representation learning is heavily driven by advanced deep generative AI models such as diffusion models (Ho et al., 2020) and variational autoencoders (Kingma and Welling, 2022) often hide exact information dynamics behind complex, non-linear transformations. To study predictive uncertainty and information gain in a fully controlled and tractable environment, we use probabilistic PCS (PPCA) from Tipping and Bishop (1999) as our analytical sandbox.

In this setup, each centered MNIST image is represented as a vector $x_n \in \mathbb{R}^D$ with $D = 28 \times 28 = 784$. The underlying generative model assumes a low-dimensional latent variable $z_n \sim \mathcal{N}(0, I_K)$ is mapped to the observed pixels via a linear Gaussian transformation, such that $x_n \mid z_n \sim \mathcal{N}(\mathbf{W}z_n, \sigma_n^2 I_D)$, where $\mathbf{W} \in \mathbb{R}^{D \times K}$ is the matrix projecting the latent coordinates back into the observed pixel space. Marginalizing out the latent variables yields a Gaussian image distribution $x_n \sim \mathcal{N}(0, S)$, where the covariance matrix $S = \mathbf{W}\mathbf{W}^\top + \sigma_n^2 I_D$ captures the learned statistical structure. Unlike deterministic architectures, PPCA learns latent representations strictly through this low-rank covariance model, naturally capturing the primary directions of spatial variation without relying on explicit encoder networks.

For masked prediction, we split the image coordinates into two complementary sets: the observed context pixels, $\mathcal{C}$, and the masked target pixels, $\mathcal{M}$. We can evaluate the model by its capacity to predict $x_\mathcal{M}$ given $x_\mathcal{C}$ entirely in closed form. Because the joint PPCA distribution is Gaussian, the resulting conditional distribution is also Gaussian.

$$p_\theta(x_\mathcal{M} \mid x_\mathcal{C}) = \mathcal{N}\left(x_\mathcal{M} \mid m_{\mathcal{M}|\mathcal{C}}, S_{\mathcal{M}|\mathcal{C}}\right), \tag{5.2}$$

where the predictive parameters are given by:

$$m_{\mathcal{M}|\mathcal{C}} = S_{\mathcal{M}\mathcal{C}} S_{\mathcal{C}\mathcal{C}}^{-1} x_\mathcal{C}, \qquad S_{\mathcal{M}|\mathcal{C}} = S_{\mathcal{M}\mathcal{M}} - S_{\mathcal{M}\mathcal{C}} S_{\mathcal{C}\mathcal{C}}^{-1} S_{\mathcal{C}\mathcal{M}}. \tag{5.3}$$

This conditional formulation turns masked prediction into a probabilistic process: the conditional mean $m_{\mathcal{M}|\mathcal{C}}$ reconstructs the missing pixels, while the conditional covariance $S_{\mathcal{M}|\mathcal{C}}$ quantifies the residual predictive uncertainty. This allows an information-theoretic data valuation, where a context region is useful if observing it reduces uncertainty about the masked data.

Predictive information gain in PPCA. We now use the PPCA predictive distribution to score candidate patches. Let Ω denote the set of all image pixels. We write $\mathcal{C}$ for the current context, $\mathcal{B}$ for a candidate patch to add to the context, and $\mathcal{M}$ for the masked target pixels whose uncertainty we want to reduce. Equivalently, we denote the corresponding random variables as

$$X_{\text{ctx}} := X_\mathcal{C}, \qquad X_{\text{patch}} := X_\mathcal{B}, \qquad X_{\text{mask}} := X_\mathcal{M}.$$

The predictive information gain of the candidate patch is defined as the reduction in predictive entropy over the masked variables:

$$\text{PredIG}_{X_{\text{mask}}}(\mathcal{B} \mid \mathcal{C}) = H(X_{\text{mask}} \mid X_{\text{ctx}}) - H(X_{\text{mask}} \mid X_{\text{ctx}}, X_{\text{patch}}). \tag{5.4}$$

Here, the first term $H(X_{\text{mask}} \mid X_{\text{ctx}})$ represents the uncertainty (entropy) about the masked region given only the current context, whereas the second term $H(X_{\text{mask}} \mid X_{\text{ctx}}, X_{\text{patch}})$ denotes the remaining uncertainty after additionally observing the candidate patch. Consequently, this quantity measures how much conditioning on the candidate patch is expected to reduce uncertainty over the masked target region.

Under PPCA, the image covariance is

$$S = WW^\top + \sigma^2 I_D. \tag{5.5}$$

For an observed set $\mathcal{O}$, the conditional covariance over the masked variables is

$$\Sigma_{\text{mask}|\mathcal{O}} = S_{\mathcal{M}\mathcal{M}} - S_{\mathcal{M}\mathcal{O}} S_{\mathcal{O}\mathcal{O}}^{-1} S_{\mathcal{O}\mathcal{M}}, \qquad \mathcal{O} \in \{\mathcal{C}, \mathcal{C} \cup \mathcal{B}\}. \tag{5.6}$$

Thus, the same covariance formula gives the predictive uncertainty before and after adding the candidate patch to the context.

Since the conditional distributions are Gaussian, the exact joint PredIG is

$$\text{PredIG}^{\text{joint}}_{X_{\text{mask}}}(\mathcal{B} \mid \mathcal{C}) = \frac{1}{2}\left[\log|\Sigma_{\text{mask}|\mathcal{C}}| - \log|\Sigma_{\text{mask}|\mathcal{C}\cup\mathcal{B}}|\right]. \tag{5.7}$$

This is the same entropy-reduction principle used in the BLR experiment, but applied to image coordinates instead of future scalar predictions.

In practice, the masked region can contain many pixels, making the joint determinant expensive and less interpretable spatially. We therefore use a diagonal pixel-wise approximation:

$$\text{PredIG}^{\text{diag}}_{X_{\text{mask}}}(\mathcal{B} \mid \mathcal{C}) = \frac{1}{2}\sum_{j\in\mathcal{M}} \log \frac{\text{Var}_*(X_{j,\theta} \mid X_{\mathcal{C},\theta})}{\text{Var}_*(X_{j,\theta} \mid X_{\mathcal{C}\cup\mathcal{B},\theta})}. \tag{5.8}$$

Here, $\text{Var}_*(X_{j,\theta} \mid X_{\mathcal{C}\cup\mathcal{B},\theta})$ denotes the predictive variance of pixel j under the fitted PPCA model, not a variance over parameters.This approximation sums marginal uncertainty reductions over masked pixels, rather than computing the full joint entropy reduction. It ignores residual correlations inside the masked region, but gives a tractable and spatially interpretable score for each patch.

A high score means that the candidate patch is informative under the PPCA covariance. Importantly, in Gaussian PPCA this score depends only on the learned covariance S and on the pixel sets $(\mathcal{C}, \mathcal{B}, \mathcal{M})$. The actual pixel values affect the conditional mean, and therefore the reconstruction, but not the conditional covariance. Thus, PPCA-PredIG provides a clean covariance-based accept–reject rule, although it is not content-adaptive.

Patch accept–reject policy. We use the diagonal PPCA-PredIG score as a gating mechanism. In the static setting, the candidate patch $\mathcal{B}$ becomes the context $\mathcal{C}$, and a random subset of pixels $\mathcal{M} \subset \Omega \setminus \mathcal{C}$ is sampled as the masked target region.

Algorithm 2 Static PPCA-PredIG Patch Accept–Reject Criterion

Require: Learned PPCA covariance S, candidate patch $\mathcal{B}$, threshold τ measured in nats, target size $M_{\text{target}} = 200$

1: Set $\mathcal{C} \leftarrow \mathcal{B}$

2: Sample $\mathcal{M} \subset \Omega \setminus \mathcal{C}$ with $|\mathcal{M}| = M_{\text{target}}$

3: Compute conditional covariance

$$S_{\mathcal{M}|\mathcal{C}} = S_{\mathcal{M}\mathcal{M}} - S_{\mathcal{M}\mathcal{C}} S_{\mathcal{C}\mathcal{C}}^{-1} S_{\mathcal{C}\mathcal{M}}$$

4: Compute score

$$s(\mathcal{B}) = \frac{1}{2}\sum_{j\in\mathcal{M}} \log \frac{\text{Var}_\theta(X_j)}{\text{Var}_\theta(X_j \mid X_{\mathcal{C}})}$$

5: Accept $\mathcal{B}$ if $s(\mathcal{B}) > \tau$; otherwise reject it.

5.1 Empirical results: PPCA, MPT and predictive filtering

We now evaluate the PPCA-PredIG framework introduced above in a controlled masked-prediction setting. The experiments address three questions. First, can PPCA be trained through masked conditional prediction? Second, how does the spatial predictive information encoded by the model evolve during MPT? Third, what happens when PredIG is used as a simple pre-update rule to bias the distribution of masked-prediction tasks? The filtering rule is used only as an illustrative application of the operator. It is not presented as a complete or optimal masking algorithm. Its purpose is to show that the covariance-based PredIG score, measured in nats, can be converted into a pre-update decision for candidate patch tasks.

All experiments use centered MNIST images represented as vectors $x \in \mathbb{R}^{784}$. We use PPCA with latent dimension $K = 20$. The noise variance σ^2 is fixed to the closed-form PPCA-ML value, and masked-SGD optimizes only the loading matrix $\mathbf{W}$. Unless stated otherwise, MPT uses target masks of 200 pixels, their 584-pixel complement as context, batch size 64, and 2000 optimization steps. Multi-run results report mean $\pm$ standard deviation over five seeds. For PredIG visualizations and filtering, candidate context regions are 10×10 image patches. The score is the diagonal pixel-wise PPCA-PredIG defined in Section 5.

Masked predictive setup. Figure 9 summarizes the masked predictive protocol used in the experiments. In our notation, $\mathcal{B}$ denotes a candidate patch, $\mathcal{C}$ the observed context, and $\mathcal{M}$ the masked target pixels. In this static PPCA setting, the candidate patch itself becomes the context, so that $\mathcal{C} = \mathcal{B}$, while $\mathcal{M} \subset \Omega \setminus \mathcal{C}$ is a set of 200 target pixels sampled outside the patch. The yellow pixels are shown only to visualize this target set: they are not provided as context to the model, but indicate the pixels over which predictive uncertainty is evaluated across permutations of the batch. The score then measures how much observing the 10×10 patch is expected to reduce predictive uncertainty over the masked targets.

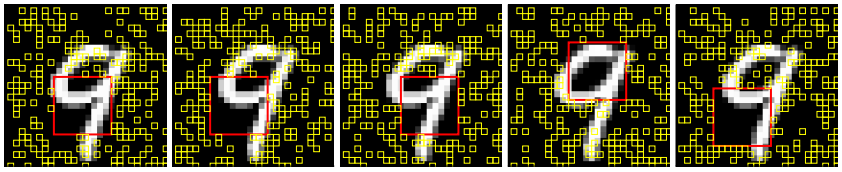


Figure 9: **Masked predictive examples for PPCA on MNIST.** Panels show the PredIG patch-selection protocol: the red square is a candidate 10×10 context patch and the yellow pixels are the masked targets.

Masked-SGD approaches the PPCA predictive reference. We first test whether PPCA can be trained through masked conditional prediction. Figure 10 compares the masked-SGD trajectory with the closed-form PPCA-ML reference evaluated on the same masked-prediction objective. The MPT score improves consistently from random initialization and approaches the predictive score obtained by the frozen PPCA-ML solution. This shows that, in this controlled linear-Gaussian setting, masked conditional prediction provides a viable training mechanism for PPCA.

PPCA-PredIG produces a spatial information map. Given a fitted PPCA covariance, every candidate patch can be scored by the expected reduction in predictive uncertainty over a target set of masked pixels. Figure 11 visualizes this computation for a representative image. The predictive mean depends on the observed context values and therefore already gives a fairly clear indication of the underlying object. In that sense, the conditional mean behaves as an image-specific reconstruction map. By contrast, the predictive standard deviation, entropy, and PredIG maps exhibit a different structure: they are driven by the conditional covariance, and therefore reflect where uncertainty remains and where uncertainty can be reduced under the PPCA model.

This distinction is informative. While the predictive mean is relatively content-specific, the variance-based maps are smoother and more spatially distributed, often forming a coarse envelope around the object rather than reconstructing its exact appearance. This happens because, under Gaussian PPCA, the conditional variance depends only on the learned covariance and on the geometry of the selected coordinates, not on the observed pixel intensities. As a result, the entropy and PredIG maps are not

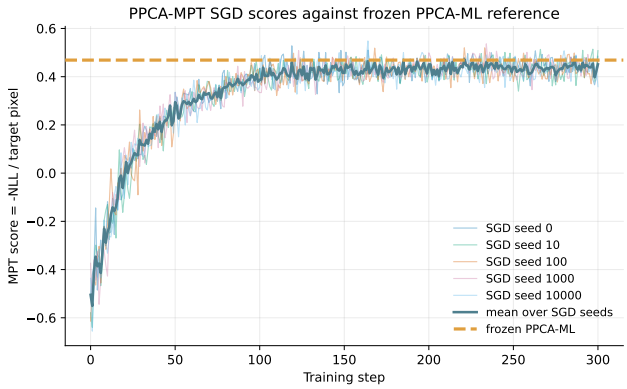


Figure 10: **Masked-SGD/MPT compared with the PPCA-ML reference.** Light curves show individual seeds, the solid curve is the mean. The dashed line is the frozen closed-form PPCA-ML model evaluated on the same centered MNIST subset and the same target/context permutations.

content-adaptive in a semantic sense, but they still reveal the spatial dependency structure captured by the model.

A second interesting pattern is that high predictive information gain often appears not only near the observed patch, where local correlations are strongest, but also in more distant regions of the image. This indicates that the learned PPCA covariance encodes non-local statistical couplings: observing one patch may reduce uncertainty both in its immediate neighborhood and in other spatially related regions. This suggests that even a simple covariance model can assign value to patches with long-range predictive influence. At the same time, it also clarifies the limitation of PPCA-PredIG: the score is spatial and covariance-driven rather than truly semantic, so it tends to follow the average support and dependency structure of the dataset rather than the exact content of each individual image.

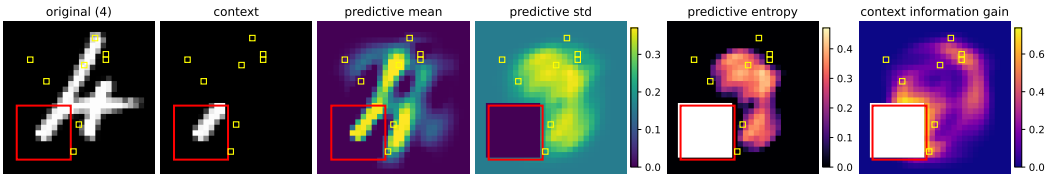


Figure 11: **Conditional uncertainty and information gain under PPCA.** PPCA gives a Gaussian conditional distribution over masked pixels. The predictive mean uses the observed context values, while the entropy and PredIG maps are determined by the covariance structure learned from MNIST and by the selected pixel coordinates.

Qualitatively similar behaviour is also observed on FASHION-MNIST and EMNIST, where the predictive mean is already highly indicative of the object class, while the variance-based maps emphasize the broader covariance structure learned from the dataset; additional examples are provided in Figure 12 and Appendix B.

PredIG reveals how predictive structure emerges during MPT. Having established that PPCA can be trained through masked conditional prediction, we next examine how the predictive information encoded by the model evolves during optimization. We freeze a collection of probe tasks, each defined by an empty initial context $\mathcal{C} = \emptyset$, a candidate 10×10 context patch $\mathcal{B}$, and a target set $\mathcal{M}$ containing 200 pixels disjoint from the patch.

For every checkpoint $\theta_t = \{\mathbf{W}_t\}$ and target pixel $j \in \mathcal{M}$, we compute the marginal pixel-wise PredIG

$$g_{t,j}(\mathcal{B}) = \frac{1}{2} \log \frac{\mathrm{Var}_{\theta_t}(X_j)}{\mathrm{Var}_{\theta_t}(X_j \mid X_{\mathcal{B}})}. \tag{5.9}$$

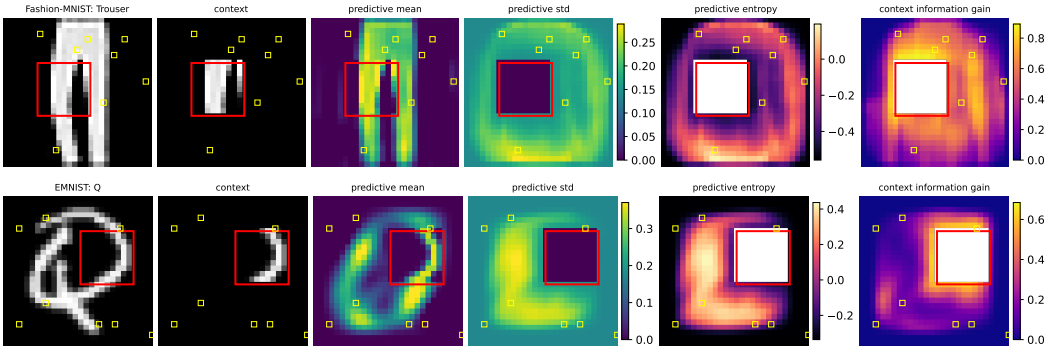


Figure 12: **Qualitative PPCA-PredIG maps on FASHION-MNIST and EMNIST.** Each row shows an example image, the observed context patch, the predictive mean, predictive standard deviation, predictive entropy, and the resulting context information gain map. Across datasets, the predictive mean is already highly informative about the object identity, whereas the variance-based maps reveal a broader covariance-driven spatial structure. High PredIG values tend to appear near the observed patch, but also in distant regions that are statistically coupled under the learned PPCA covariance.

The patch and target coordinates are held fixed across checkpoints. Therefore, changes in $g_{t,j}$ reflect changes in the predictive covariance learned by MPT rather than changes in the masked-prediction task.

Figure 13 shows that MPT progressively modifies not only predictive accuracy, but also the spatial information structure of the model. At initialization, the loading matrix $\mathbf{W}_0$ is small and random, so the PPCA covariance is close to a diagonal noise covariance. As a result, cross-covariances between the candidate patch and the target pixels are weak, observing $\mathcal{B}$ produces little reduction in target variance, and PredIG is close to zero. During MPT, $\mathbf{W}_t$ is optimized through repeated masked conditional prediction tasks.

The bright pixels in Figure 13 should therefore be interpreted as locations of high uncertainty reduction. They are not simply pixels with high predictive variance. A target pixel has high PredIG only when a substantial part of its uncertainty can be removed by observing the particular candidate patch $\mathcal{B}$. This also explains why high values may appear both near the patch and in more distant regions.

The MNIST images are included as spatial references. Under Gaussian PPCA, however, PredIG depends on the learned covariance and on the geometry of $(\mathcal{B}, \mathcal{M})$, but not on the observed pixel intensities. The maps should therefore be interpreted as covariance-based predictive dependencies rather than image-specific semantic explanations.

From predictive information to patch filtering. The previous experiment uses PredIG as a diagnostic operator. The same score can also be used operationally. We aggregate the pixel-wise values over the target set,

$$s_t(\mathcal{B}) = \sum_{j \in \mathcal{M}} g_{t,j}(\mathcal{B}), \tag{5.10}$$

and apply the threshold rule defined in Section 5. A candidate patch task is accepted when its score is above the information threshold τ and rejected otherwise.

Figure 14 illustrates this rule across candidate patch tasks. The goal of this experiment is to show that PredIG can be converted into a simple pre-update - online selection decision.

Information-based filtering induces task specialization. Using PredIG for filtering changes the distribution of masked-prediction tasks seen during training. Moreover, because PPCA-PredIG encodes spatial information through the learned covariance structure, we evaluate each model on two complementary validation distributions: uniformly sampled random masks and high-PredIG masks. High-PredIG masks are defined as candidate patches whose PPCA-PredIG score lies above the 0.60 quantile. This comparison does not test whether PredIG is universally better than random masking.

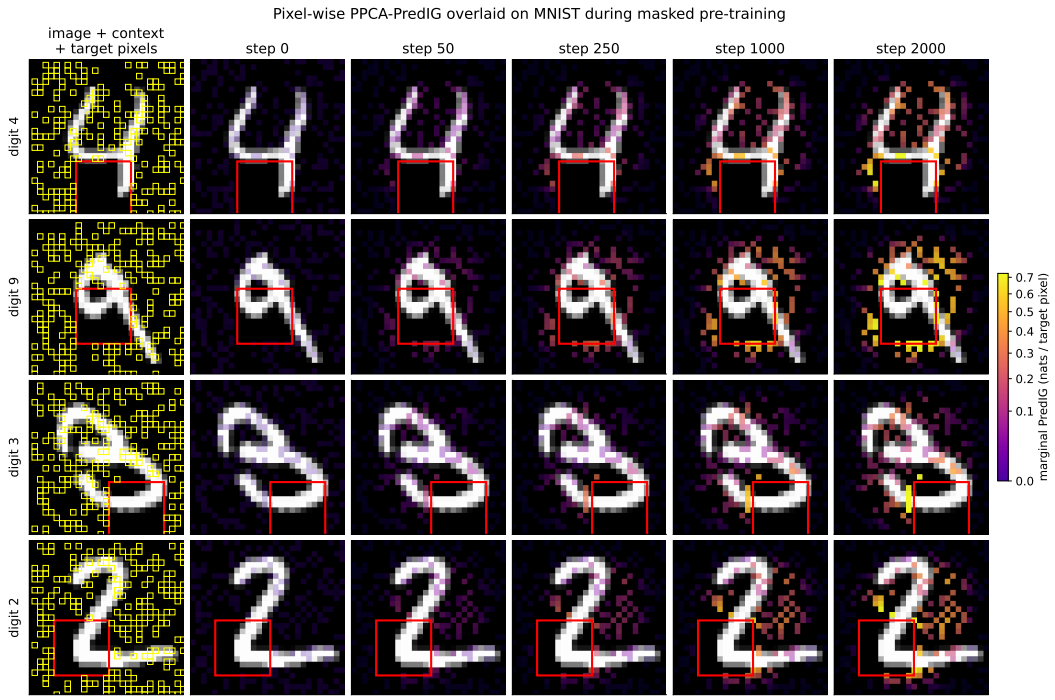


Figure 13: **Evolution of pixel-wise predictive information during PPCA-MPT.** The first column defines each fixed probe task. The red square denotes the candidate 10×10 context patch $\mathcal{B}$, while the yellow squares identify the 200 target pixels in $\mathcal{M}$. The remaining columns show the marginal pixel-wise PredIG at different MPT checkpoints, overlaid on the corresponding MNIST image. The patch, target set, and color normalization remain fixed across checkpoints. At initialization, predictive information is weak and largely unstructured. As masked training progresses, PredIG becomes larger and spatially organized, revealing the predictive dependencies encoded by the learned PPCA covariance.

Instead, it measures the trade-off between broad mask coverage and specialization toward the tasks prioritized by the information operator.

Table 3 shows the resulting dual validation. On random masks, strong PredIG bias hurts performance. The random policy and the weakly biased 25% PredIG mixture are similar, but the 75% and 100% PredIG policies degrade random-mask NLL and MSE. On high-PredIG masks, the pattern reverses: performance improves as the amount of PredIG sampling increases, and the 100% PredIG policy obtains the best high-PredIG validation NLL and MSE. PredIG therefore induces specialization rather than a universal improvement.

Table 3: **Dual-validation performance.** Results are mean $\pm$ standard deviation over five seeds. Lower NLL and MSE are better. The covariance error is the relative Frobenius error to the closed-form PPCA-ML covariance.

Validation split	Policy	NLL $\downarrow$	MSE $\downarrow$	Cov. error to ML $\downarrow$
Random masks	Random	-0.286 ± 0.010	0.0704 ± 0.0016	1.949 ± 0.044
	25% PredIG	-0.288 ± 0.004	0.0706 ± 0.0010	1.936 ± 0.048
	50% PredIG	-0.286 ± 0.009	0.0710 ± 0.0012	1.871 ± 0.044
	75% PredIG	-0.278 ± 0.008	0.0732 ± 0.0006	1.891 ± 0.053
	100% PredIG	-0.262 ± 0.008	0.0771 ± 0.0014	2.002 ± 0.048
High-PredIG masks	Random	-0.322 ± 0.010	0.0606 ± 0.0015	1.949 ± 0.044
	25% PredIG	-0.343 ± 0.009	0.0587 ± 0.0007	1.936 ± 0.048
	50% PredIG	-0.350 ± 0.004	0.0586 ± 0.0005	1.871 ± 0.044
	75% PredIG	-0.362 ± 0.011	0.0578 ± 0.0011	1.891 ± 0.053
	100% PredIG	-0.373 ± 0.015	0.0575 ± 0.0016	2.002 ± 0.048



Figure 14: **Illustrative PPCA-PredIG accept–reject rule.** Each candidate 10×10 patch is scored by the aggregated pixel-wise PredIG over the target pixels. High-scoring patch tasks are accepted and low-scoring patch tasks are rejected according to a fixed threshold τ. Under Gaussian PPCA, this decision depends on the covariance structure and the geometry of the masked-prediction task, not on the semantic content of the image.

Choice of mask size. In the main experiments, we fix the target and context sizes to isolate the effect of PredIG-based task selection. We additionally perform a sweep over different mask sizes, reported in Appendix C.1. This diagnostic confirms that the raw MPT objective is not directly comparable across mask sizes, since it sums conditional log-probabilities over different numbers of target pixels. For this reason, cross-mask comparisons are reported per target pixel. The sweep shows that PPCA-MPT approaches the frozen PPCA-ML predictive reference across context sizes, while larger observed contexts generally yield better normalized predictive scores.

6 Future directions and discussion

While the previous sections established the analytical foundations of predictive information gain using closed-form linear-Gaussian models, scaling this framework requires moving beyond rigid spatial covariances. In this section, we first present a preliminary empirical evaluation using deep probabilistic ensembles to demonstrate how the accept–reject principle transitions to content-adaptive uncertainty. We then discuss the broader algorithmic, theoretical, and computational implications of these findings, outlining key pathways for scaling predictive data valuation to modern representation learning systems.

6.1 Deep ensembles

From spatial covariance to content-adaptive uncertainty. The PPCA experiment exposes a limitation of the linear-Gaussian setting. In PPCA, the conditional covariance depends only on the learned covariance matrix S and on the selected pixel sets. It does not depend on the actual observed pixel values. As a result, PPCA-PredIG is mainly a spatial score: it tells us which patch locations are informative under the global MNIST covariance, but not whether the content of a particular patch is informative for the current image.

This makes PPCA useful as a closed-form *proof of concept* for predictive filtering, but insufficient as a content-adaptive admission rule. To make the decision image-dependent, the model must output a predictive distribution, not only a point reconstruction. We therefore replace PPCA with a deep probabilistic ensemble trained on the same masked-prediction task, following the use of ensembles as practical uncertainty estimators Lakshminarayanan et al. (2017).

Given an image x and a candidate context patch B, the ensemble predicts the target pixels $M(B)$ and returns both a conditional mean and a predictive variance. We summarize the remaining uncertainty by the conditional predictive entropy

$$u(B; x) = H_\theta\big(X_{M(B)} \mid X_B = x_B\big)\,, \tag{6.1}$$

and define the accept–reject score as

$$s(B; x) = -u(B; x). \tag{6.2}$$

Thus, high-scoring patches are those after which the model expects low remaining uncertainty about the missing pixels. The accept–reject principle is unchanged: admit candidates that make the target easier to predict. The difference is that the score is now conditioned on the observed visual content, rather than fixed by spatial covariance alone.

Ensemble accept–reject results. We evaluate the ensemble score offline on 800 candidate patches from 100 MNIST images. A fixed quantile threshold accepts 329 candidates and rejects 471. We then compare the actual predictive quality of accepted and rejected candidates.

Accepted patches lead to better missing-pixel prediction. Their mean MSE is 0.0581, compared with 0.0661 for rejected patches. The same pattern appears in negative log-likelihood: accepted patches obtain a mean NLL of -1.8164, compared with -1.7107 for rejected patches. Since lower MSE and lower NLL are better, the ensemble score is aligned with downstream predictive quality.

Table 4: Offline accept–reject validation using the deep probabilistic ensemble. Lower MSE and NLL are better.

Metric	Accepted	Rejected	Rejected – Accepted
Number of candidates	329	471	–
MSE	0.0581	0.0661	0.0080
NLL	-1.8164	-1.7107	0.1057

We finally test whether the ensemble score remains mainly location-driven, as in PPCA, or whether it also reacts to the specific image content. This is important because the main limitation of PPCA-PredIG is that its conditional covariance, and therefore its entropy map, is fixed by the learned dataset-level covariance and does not depend on the observed pixel values of the particular image.

To probe this effect, we compare the score grid of an original image with the score grid obtained after shifting the same image. The two grids have correlation (0.8434), mean absolute difference (0.0345), and maximum absolute difference (0.0868). The high correlation indicates that the ensemble score still contains a strong spatial component, which is expected on MNIST, where digit structure is strongly aligned with image position. However, the non-zero differences show that the uncertainty estimate is not determined by location alone: it also changes when the observed content changes.

Interpretation. The PPCA and ensemble experiments therefore play complementary roles. PPCA provides the analytic mechanism: masked conditional prediction, entropy maps, predictive-information maps, and accept–reject filtering can all be computed explicitly. Its limitation is that the score is mostly geometry-driven, because the conditional covariance is independent of the observed pixels.

The ensemble experiment addresses this limitation by introducing content-adaptive predictive uncertainty. With this richer uncertainty model, the information score is still influenced by spatial regularities, but it is no longer fixed by location alone. This makes the accept–reject criterion more expressive: accepted patches have lower reconstruction error and better likelihood than rejected patches. The empirical conclusion is therefore not that PPCA alone solves predictive filtering, but that the predictive-information principle becomes practically useful when the uncertainty model is expressive enough to react to image content.

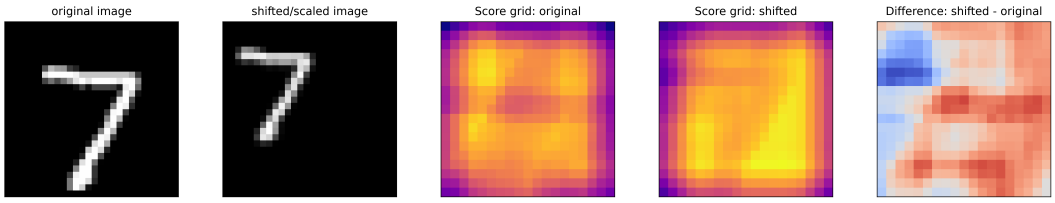


Figure 15: **Content-adaptive uncertainty from the probabilistic ensemble.** The ensemble assigns entropy-based scores to candidate context patches. Unlike PPCA-PredIG, these scores depend on the observed image content, not only on patch geometry.

6.2 Limitations and open challenges

We provide a first step towards predictive data valuation from an information-theoretic perspective. The central contribution is not a final data-processing algorithm, but a controlled study of how predictive information gain behaves in probabilistic models where uncertainty can be explicitly computed. The next step is to move from this analytical setting towards richer models, larger datasets, and practical training protocols.

A first limitation comes from the PPCA setting. PPCA makes the operator analytically clean because the predictive distribution is Gaussian and the conditional covariance is available in closed form. However, this also imposes an important restriction: the PredIG score is driven by the learned covariance and by the spatial configuration of the observed and masked pixels. It is therefore not content-adaptive in a semantic sense. The predictive mean may already reveal the object structure in an image, but the uncertainty, entropy, and PredIG maps are governed by the variance structure learned at the dataset level. Future work should study whether the same operator can become more image-adaptive when implemented with richer non-linear models.

A natural direction is to extend the analysis to deep probabilistic predictors. The preliminary Deep Ensemble experiments open this path by replacing the linear-Gaussian covariance of PPCA with input-dependent predictive uncertainty. Larger models or different architectures would make possible to test whether PredIG can capture not only spatial covariance patterns, but also content-dependent uncertainty in masked representation learning and, therefore, be evaluated in different scenarios, from text, pixels or latent representations.

A second direction is algorithmic. In this project, accept–reject rules are used as simple diagnostic mechanisms to test whether predictive information can be converted into a decision. Future work should develop optimized protocols for using PredIG during training. This includes studying how thresholds should be chosen, whether they should adapt during learning, how candidate masks should be generated, and how the cost of computing the information score compares to the cost saved by avoiding redundant updates. The goal would be to turn the operator from a diagnostic quantity into a practical mechanism for efficient data assimilation.

Finally, the computational implications should be quantified explicitly. If a predictive information criterion reduces redundant processing, its benefit should be measured not only in terms of predictive performance, but also in terms offloating-point operations, GPU hours, energy consumption, and training cost. This is especially relevant for large-scale self-supervised learning, where high performance is often obtained by repeated passes over massive datasets. A full future study should therefore connect predictive information gain with concrete efficiency metrics, asking how many updates can be avoided, how much computation is saved, and under what conditions predictive performance is preserved.

Overall, future work should investigate whether the principles studied here in controlled Bayesian and PPCA models can be transferred to modern representation-learning systems. The long-term goal is to understand whether training can be guided not only by the amount of data available, but by the amount of predictive information that data is expected to provide.

7 Conclusions

This work has studied whether all masked prediction tasks are equally useful for learning. To address this question, we introduced predictive information gain (PredIG), an information-theoretic operator that measures the expected reduction in predictive uncertainty produced by a candidate observation or batch. The central idea is to value data by its effect on the model's observable predictions, rather than by the amount of data processed or by changes in parameter uncertainty.

We first analysed PredIG in Bayesian linear regression, where the posterior and predictive distributions can be computed in closed form. This setting made it possible to inspect the operator sequentially and showed that predictive information is state-dependent: observations become redundant when the model can already predict the corresponding region, while informative regions can reappear as the predictive state changes. We then extended the analysis to PPCA, using it as a tractable probabilistic model for masked image prediction. In this setting, PredIG produces spatial information maps and provides a closed-form way to score candidate context patches.

The empirical results support the main hypothesis of the project: masked observations are not equally informative. A PredIG-based accept–reject policy can identify redundant observations and reduce the amount of data assimilated during training while preserving predictive performance. In selected BLR experiments, the policy rejects more than 80% of candidate observations with little degradation in the final predictive metrics. This suggests that predictive information can act as a useful criterion for deciding when an update is worth performing.

At the same time, the work also clarifies the limits of the current setting. In PPCA, PredIG is driven by the learned covariance structure and is therefore mainly spatial rather than semantic or content-adaptive. The results should therefore be understood as an analytical *proof of concept*, not as a final scalable training algorithm. Nevertheless, they show that information-theoretic data valuation can be made explicit, computed in closed form, and connected to masked prediction.

Overall, we provide a principled step towards viewing self-supervised learning not only as a problem of exposing models to more data, but as a problem of identifying which observations provide predictive information. The main conclusion is that redundancy can be quantified, and that masked prediction tasks can be valued by the uncertainty they remove.

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A Conditional entropy and predictive information gain

Information theory measures uncertainty through entropy. For a continuous random variable X with density $p(x)$, the differential entropy is defined as

$$H(x) = -\int p(x) \log p(x)\, dx. \tag{A.1}$$

When another random variable Y is observed, the remaining uncertainty about X is measured by the conditional differential entropy

$$H(x \mid y) = -\int p(x, y) \log p(x \mid y)\, dx\, dy. \tag{A.2}$$

Equivalently, this can be written as the expected entropy of the conditional distribution:

$$H(x \mid y) = \mathbb{E}_{p(y)} \left[H(x \mid y_i) \right]. \tag{A.3}$$

Thus, conditional entropy measures the uncertainty that remains about x after observing y. The reduction in uncertainty induced by observing Y is the mutual information

$$I(x; y) = H(x) - H(x \mid y), \tag{A.4}$$

which is always non-negative (Cover and Thomas, 2006).

In the predictive setting considered in this work, the object of interest is not the uncertainty over parameters, but the uncertainty over a fixed set of target predictions. Let x_* be a fixed collection of target inputs and Y_* the corresponding predictive variables. Since x_* remains fixed, we omit it from the notation below.

For a single candidate input x, PredIG is the reduction in predictive uncertainty induced by x:

$$\mathrm{PredIG}(x) = \underbrace{H_{\mathrm{before}}[y_*]}_{\text{before } x} - \underbrace{H_{\mathrm{after}}[y_* | x]}_{\text{after } x}, \tag{A.5}$$

This is precisely the conditional mutual information between the target predictions and the candidate observation:

$$\mathrm{PredIG}(x) = I(y_*; y_x \mid \mathcal{D}_N). \tag{A.6}$$

B Additional PPCA-PredIG examples

Figure 16 provides additional qualitative examples on FASHION-MNIST and EMNIST. As in the MNIST case, the predictive mean is already strongly indicative of the object of interest, whereas the predictive standard deviation, predictive entropy, and PredIG maps follow the covariance structure learned by PPCA. This often produces a broader object-shaped envelope, with high information values both near the observed patch and in more distant but statistically dependent regions.

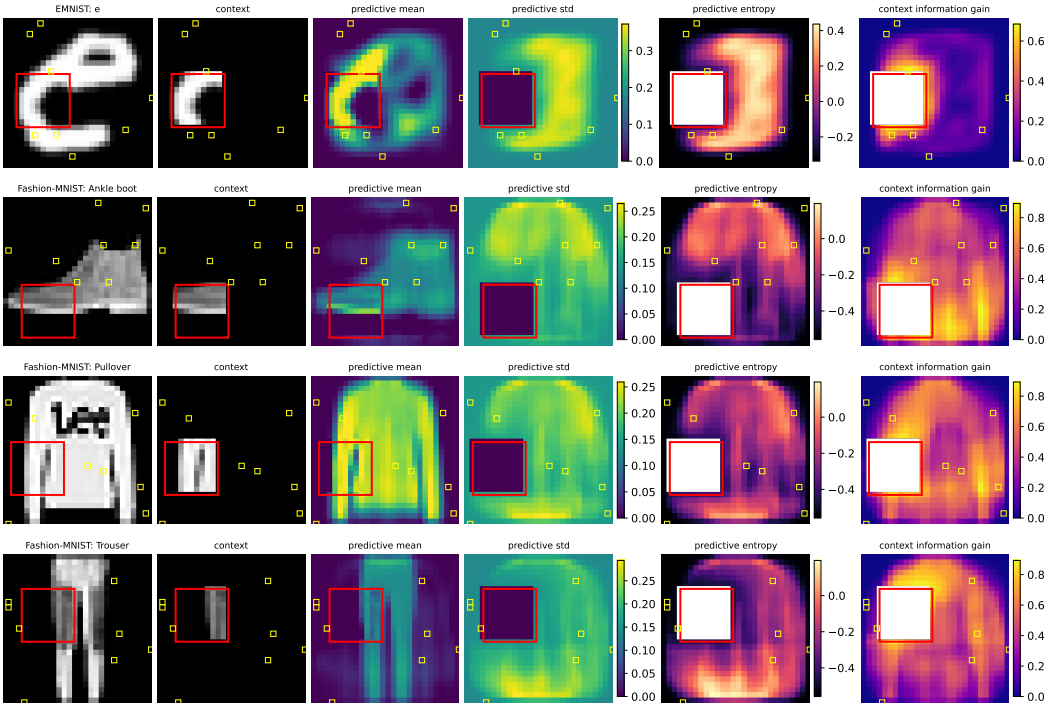


Figure 16: **Additional PPCA-PredIG qualitative examples on FASHION-MNIST and EMNIST.** Rows illustrate examples from FASHION-MNIST and EMNIST. The predictive mean is image-specific and already provides a clear cue about the object, while the predictive standard deviation, entropy, and context information gain maps are governed by the learned covariance structure. These maps therefore reflect spatial predictive dependencies rather than content-adaptive semantic reasoning.

C Additional PPCA-MPT diagnostics

C.1 Effect of context size on PPCA-MPT convergence

We include an additional diagnostic experiment to assess how the size of the observed context affects PPCA-MPT convergence. This experiment is not part of the main PredIG filtering result. Its purpose is to verify that the masked predictive objective behaves consistently across masking regimes, and to clarify the effect of context size on optimization.

For an image with $D = 784$ pixels, we vary the number of observed context pixels $|R|$ and masked target pixels $|M|$, with $|R| + |M| = D$. We consider

$$|R| \in \{764, 734, 684, 584, 384, 184, 56\}, \qquad |M| \in \{20, 50, 100, 200, 400, 600, 728\}.$$

For each setting, PPCA-MPT is trained from random initialization and compared with the frozen closed-form PPCA-ML solution evaluated under the same masking regime. Results are averaged over five random seeds.

Since the masked predictive objective sums conditional log-probabilities over the target pixels, raw values are not directly comparable across different target sizes. We therefore report the normalized masked predictive score

$$\mathcal{S}_\theta(M, R) = \frac{1}{|M|} \log p\theta(x_M \mid x_R),$$

which measures the average masked predictive log-probability per target pixel.

Figure 17 shows that PPCA-MPT improves consistently across all context sizes and approaches the PPCA-ML reference. The main qualitative effect is that smaller contexts make the masked prediction problem harder. When only a small number of pixels is observed, the model has less conditional information available, and convergence becomes slower. This is most visible in the extreme setting

with $|R| = 56$ and $|M| = 728$, where the curve still improves steadily but approaches the reference more gradually.

Overall, the diagnostic supports the use of PPCA-MPT as a stable masked-prediction testbed. The convergence behaviour is consistent across masking regimes, while confirming the expected role of context: larger observed contexts provide more information and make conditional prediction easier.

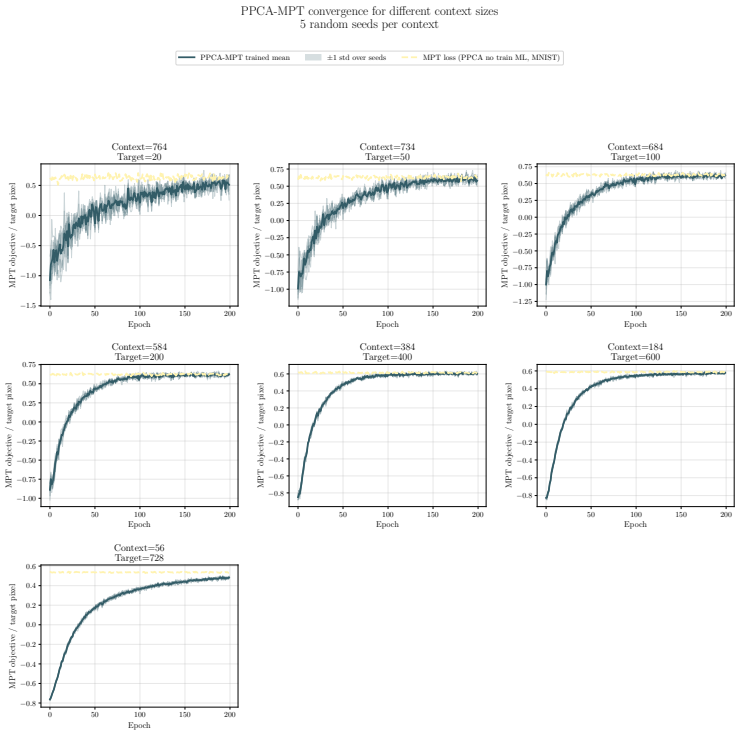


Figure 17: **Effect of context size on PPCA-MPT convergence.** Each panel shows the normalized MPT objective per target pixel for a different context–target split on MNIST. The solid curve is the mean PPCA-MPT trajectory over five random seeds, the shaded region denotes ± 1 standard deviation, and the dashed curve is the frozen closed-form PPCA-ML reference evaluated under the same masking regime. PPCA-MPT converges across all context sizes, but the smallest-context regime converges more slowly, indicating that masked prediction becomes harder when less context is available.